

14 Local Vibration Effects on Range of Motion and Performance

Effects on Range of Motion (ROM)

Local muscle vibration (see Figure 14.1) alone and combined with static stretching have been used to enhance range of motion (ROM) improvements. Vibration (35 Hz with 2 mm amplitude displacement) and static stretching have increased hamstrings flexibility more than static stretching (7.8 percent) alone (10), while the combination has also improved the ability to perform the forward splits with male gymnasts (19). With synchronized swimmers, vibration (30 Hz at 2 mm displacement) improved passive but not active ROM (forward splits) (18). Local vibration (30 Hz at 4 mm displacement) alone demonstrated similar stand and reach ROM (lower back and



Figure 14.1 Local vibration device

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hamstrings) increases as static stretching with better results than dynamic stretching and whole body vibration (14). There were no statistically significant differences compared to static stretching but local vibration-induced ROM was consistently higher with moderate effect size magnitudes. The vibration effects were especially effective with the most flexible participants (14). An early study from 1976 used 15 minutes of vibration to the low back and hamstrings (44 Hz with 0.1-mm displacements) and reported similar increases in ROM as static stretching (1). Their vibration displacement was very small which could have been the difference between their study and other studies that did show a greater benefit with vibration. Two studies vibrating the plantar flexors found a large magnitude increase in dorsiflexion ROM with no changes in plantar flexor MVC torque (13) while the other study showed similar increases in ankle ROM when comparing plantar flexors static stretching to vibration and static stretching combined (15). Local vibration of the quadriceps (30 Hz vibration with 10 mm amplitude for ten seconds with five seconds recovery) increased knee ROM immediately after and ten minutes after the vibration intervention, whether male American football players had prior knee injuries or not (12).

Although five sessions of local vibration with a Theragun significantly improved active ankle dorsiflexion ROM, Instrument Assisted Soft Tissue Mobilization (IASTM) provided greater ROM increases (21). The authors suggested that IASTM penetrates the muscle deeper than Theragun local vibration providing better fibroblast and fibroblast cell formation and collagen fibres remodeling. A recent systematic review of 11 studies (7) reported that an acute session of percussive massage devices could be effective for increasing iliopsoas, hamstrings, triceps surae and posterior chain muscle flexibility.

Effects on Performance

Although, the improved flexibility with vibration is relatively consistent, research reports on its effect on subsequent performance is mixed. Vibration has counteracted static stretch-induced impairments in jump height (6), and knee flexor and extensor forces (10), in two studies but demonstrated that vibration alone or in combination with static stretching still induced impairments in strength and activation (8, 15) in two other studies. The previously cited Ferreira et al. systematic review (7) reported a lack of improvements in strength, balance, acceleration, agility, and explosive activities.

Mechanisms

The mechanisms underlying the increased ROM with vibration have been attributed to a myriad of factors (18). The factors include increased stretch (pain) threshold, increased blood flow associated with an increase in temperature or vibration stress (11, 16, 17), which would decrease muscle viscosity reducing the resistance to muscle extensibility. Increases in passive straight

leg raise ROM following 2 x 60 seconds of handheld percussive therapy (29 Hz) with 30-seconds of rest between the repetitions induced an increase in hamstrings ROM in association with a resultant decrease in tissue stiffness (20). Fifteen minutes of back percussive massage (30 Hz) did not increase lumbar (lower back) flexibility or affect thoracolumbar fascia thickness. But there were decreases in echo intensity and perceived stiffness and an elevated skin temperature (23). A decrease in echo intensity is purported to reflect muscle quality, owing to an attenuation of intramuscular fibrous and adipose tissue. The authors suggested that the decreased echo intensity may have been related to a reduction in the viscosity of hyaluronic acid between the loose connective tissue. Vibration-induced muscle relaxation (9, 22), and decreases in the phasic and static stretch reflexes (2) have also been reported. With even greater nuance, high frequency (90 Hz) vibration affected short-latency responses more than medium-latency responses, but when the vibration frequency was reduced to 30 Hz there was negligible effect on the short-latency reflex, but the medium-latency reflex was significantly reduced (4). Bove and colleagues (5) suggested that these effects were due to pre-synaptic inhibition of the group Ia afferent fibres or a gate control type limitation where both vibration and stretching influence the same Ia pathways. Furthermore, if the stress is high enough, the combination of a strong stretch stimulus and substantial vibration could activate Golgi tendon organ inhibition through the Ib pathways (autogenic inhibition). However, Golgi tendon organ inhibition only persists for less than a second after the stimulus is removed. Additionally, other factors might include intrafusal fibre fatigue or persistence of motoneuron after-discharge, owing to reverberation of the interneuron pool (3). In summary, percussive massage may elicit a number of ROM enhancing mechanisms such as the ubiquitous increase in stretch/pain tolerance, increased blood flow and tissue temperature promoting a decrease in tissue viscosity (thixotropic effects). There also seems to be an attenuation of various reflexes but the post-vibration duration of these effects would be very brief.

Summary

So, adding vibration to the stretching routine is probably not necessary for the average individual interested in improving this musculoskeletal health factor (flexibility), as stretching on its own should accomplish that objective. Alternatively, the research indicates that local vibration in many cases is as effective as stretching for improving ROM. Of course, you do not have to spend money on a percussive massage device if you just stretch. However, for those individuals who need to go further and achieve more extreme ROM, then vibration in concert with stretching might be recommended.

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